CARGOBIT SECURITY

Technical Deep-Dive

Architecture and implementation details for engineering leads: core components, data flows, reliability patterns, and observability.

CargoBit Platform

Version 1.0

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CONFIDENTIAL

1. Core Architecture

Die Core Architecture besteht aus fünf Services: Security Gateway (Decision Layer), Risk Engine (Real-Time Scoring), Mitigation Service (Adaptive Controls), Audit Service (Immutable Events) und Notification Service (Alerts).

Service	Port	Tech Stack	Scaling
Security Gateway	3004	Node.js / Express	Horizontal
Risk Engine	3003	Python / FastAPI	Horizontal
Mitigation Service	3005	Node.js / BullMQ	Worker Pool
Audit Service	3006	Go / PostgreSQL	Write-Sharding

2. Data Flows

Data Flow: Domain → Gateway → Risk → Mitigation → Audit → Notification. Correlation-IDs für Traceability, idempotente Requests, async Processing via Queues.

3. Reliability Patterns

Pattern	Implementation	Ziel
Circuit Breaker	Hystrix / resilience4j	Cascading Failures verhindern
Retries + Backoff	Exponential Backoff	Transient Errors handhaben
Horizontal Scaling	Kubernetes HPA	Traffic-Spitzen abfangen

4. Security Controls

- mTLS: Service-to-Service Encryption
- Service-JWT: 5-min expiry, auto rotation
- AES-256 at rest: Database encryption
- PFS enforced: Perfect Forward Secrecy

5. Observability

Komponente	KPI	Target
Gateway	P95 Latency	< 120 ms
Risk Engine	P95 Latency	< 80 ms
Mitigation Queue	Queue Lag	< 2s

Audit Service	Write Latency	< 50 ms